Solvency Field Theory V12.11

Master Action Audit and Upgrade:
From Macro Ledger to Sector Stack

The repaired V11 action survives;
V12.11 adds tested sector layers around it

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The Master Action is allowed to be incomplete, but it is not allowed to pretend.

Publication boundary. V12.11 audits the repaired V11 Lagrangian against all SFT advancements from V11.1 through V12.10 and produces a tested Master Action v1 stack. The repaired core action survives unchanged. V12.11 adds placement layers for maintained worldline histories, no-overwrite admissibility, support-domain constraints, frame/holonomy interface, effective mass focusing, effective thermal formation, and a weak-field morphology projection bridge. V12.11 does not derive full continuum PDEs, gauge couplings, particle maps, physical masses, or cosmogenesis closure.

Contents

1 The physical picture

The original V11 Lagrangian was written when SFT was mostly a gravitational accounting theory. Particles were operation-density sources. They ticked, accumulated boundary burden, and the ledger produced gravity and expansion. The action treated matter as an input:

Given matter clocks, here is the ledger.

V12.4 through V12.10 changed the ontology underneath the action. Particles are no longer featureless sources. They are maintained no-overwrite helical worldline structures in 3+1 spacetime. Their internal cadence, framing, support ecology, and focusing geometry generate the operation current that the old action consumes. The new picture is:

Here is what matter is, and here is why the ledger follows from it.

V12.11 asks: does the old action need to be replaced? The answer is no. It needs to be *extended.* The repaired V11 core remains the macro ledger backbone. But the source terms that feed it — $J^\mu_{\rm ops}$, $T_{\mu\nu}$, $\mathcal{L}_m$ — are no longer primitive. They are the coarse-grained output of structured particle histories. V12.11 makes that structure explicit by wrapping the old core in a sector stack.

1.1 The sector stack

The Master Action v1 is:

$$S^{12.11}_{\rm SFT} = S_{\rm core} + S_{\rm hist} + S_{\rm NOW} + S_{\rm frame} + S_{\rm support} + S^{\rm eff}_{\rm focus} + S_{\rm eff/thermal} + P_{\rm shape}/\lambda_{\rm bridge} \tag{1}$$

Each sector has an explicit certification status.

Written out explicitly, the combined action is:

$$\begin{aligned} S^{12.11}_{\rm SFT} = &\int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \eta\,\nabla_\mu\chi\,J^\mu_{\rm ops} + \Lambda_C C_{\rm min} - V_{\rm quad}(\psi) + \mathcal{L}_m - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \tfrac{\kappa_\theta}{2}\,D_\mu\theta\,D^\mu\theta\right] \\ &+ \sum_i \int d\tau_i\,\mathcal{L}_{\rm hist}[X^\mu_i, \phi_i, \nu_{C,i}] \\ &+ \int_\Sigma d^3x\sqrt{\gamma}\left[\Lambda_{\rm NOW}C_{\rm NOW} + \Lambda_{\rm supp}C_{\rm RC} + \Lambda_{\rm cell}\big(\Pi_{\rm adm} - \tfrac{3}{4}\big)\right] \\ &+ S^{\rm eff}_{\rm focus} + S_{\rm eff/thermal} + \mathcal{P}_{\text{weak-field}}[J_{\rm ops}, M_{ab}, h, \Pi_{\rm adm}, \lambda]. \end{aligned} \tag{2}$$

The first line is the original repaired V11 core plus the frame/holonomy interface. The second line is the worldline-history source bridge. The third line contains the no-overwrite, support-domain, and disk-cell constraints as Lagrange-multiplier layers. The fourth line collects the effective downstream sectors and the weak-field projection readout.

1.2 Three results from the audit

The audit produced three results that were not anticipated.

The source-force boundary result (R012). Substituting the worldline current into the direct χ-current coupling gives, for a conserved fixed-cadence history:

$$S_{\chi,i} = -\eta\nu_{C,i}\int d\tau\, u^\mu_i\partial_\mu\chi = -\eta\nu_{C,i}[\chi(X_f) - \chi(X_i)]. \tag{3}$$

This is an exact boundary term. The bookkeeping scalar does not produce a local fifth force on conserved particles. Motion comes through the settlement geometry (the metric), not through a scalar gradient. This resolves a long-standing concern about the observability of the χ field.

The disk admissibility factor (R018). The normalization constant κ in the weak-field disk projection is not a fitted parameter. It derives from the 3+1 slot binary no-overwrite disk cell constraint:

$$\kappa_{\text{disk}} = \frac{3}{4}, \tag{4}$$

connecting the galactic rotation-curve program directly to V12.7's mechanical filter theorem. The same no-overwrite structure that selects nine particle modes also constrains the disk morphology projection.

The disk endpoint (R014–R019). The interpolating function endpoint for disk-dominated galaxies is:

$$\lambda_{\text{disk}} = \frac{\pi}{2} \times \frac{3}{4} = \frac{3\pi}{8}, \tag{5}$$

where $\pi/2$ is the raw ideal disk chord and $3/4$ is the admissibility factor. The sphere endpoint remains $\lambda_{\text{sphere}} = 4/3$ as derived in V11.

2 Core survival (R002)

The repaired V11 Lagrangian survives as S_{core}:

$$S_{\text{core}} = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \eta\,\nabla_\mu\chi\,J^\mu_{\text{ops}} + \Lambda_C C_{\min} - V_{\text{quad}}(\psi) + \mathcal{L}_m\right]. \tag{6}$$

Ten hard tests. Zero failures. The matched static vacuum branch reduces to Einstein–Hilbert. The quadratic lag potential has $V(0) = 0$, $V'(0) = 0$, $V''(0) > 0$. No fake cosmological constant. The core is load-bearing and it holds.

3 Worldline-history bridge (R003, R005)

The operation current is placed as the coarse-grained output of maintained particle histories:

$$J^\mu_{\text{ops}}(x) = \sum_i \int d\tau_i\,\nu_{C,i}\,u^\mu_i(\tau_i)\,\delta^{(4)}_g(x - X_i(\tau_i)), \tag{7}$$

where $\nu_{C,i} = m_i c^2/h$ is the Compton cadence. R003 verified that a stationary particle gives the expected operation density, that boosted particles transform as a 4-current, and that many-particle coarse-graining recovers the fluid source. R005 verified that internal Compton phase accumulates at ν_C without creating a preferred frame, that the helical representation is Lorentz-compatible, and that the "helix" is internal phase holonomy, not a literal spatial corkscrew.

4 No-overwrite admissibility (R004, R018–R019)

No-overwrite is promoted from a founding principle to a constraint layer S_{NOW}. R004 established the toy admissibility rule: written support is locked, only the next write varies, one-speed cap limits candidates, least-cost flat branch preserves straight continuation, and gradients bend future writes without rewriting the past.

R018–R019 derived the disk support-cell constraint. The admissibility projector for a planar disk cell is:

$$\Pi_{\mathrm{adm,disk}} = \frac{3}{4}, \tag{8}$$

arising from the 3+1 slot binary no-overwrite structure. This was promoted into the $S_{\mathrm{NOW}}/S_{\mathrm{support}}$ layer (R019), connecting the microscopic particle-mode constraint to the macroscopic galactic morphology correction.

R024–R025 upgraded the smooth admissibility scalar from plain flattening to a support-lock tensor scalar. The no-overwrite slot table says a disk locks its normal direction while leaving the two planar tangent slots admissible. The correct question is not "how flat?" but "does a plane exist?" — measured by the ratio of the smallest to the middle eigenvalue of the local written-support tensor:

$$h_{\mathrm{planar}} = 1 - \sqrt{\frac{\sigma_3^2}{\sigma_2^2}}, \tag{9}$$

with $h = 0$ when $\sigma_2 \to 0$ (no plane exists). A current-history correction distinguishes settled disks from transient pancakes:

$$h_{\mathrm{lock}} = h_{\mathrm{planar}}\,(1 - f_\perp), \qquad f_\perp = \frac{\langle (J_{\mathrm{spatial}} \cdot \hat{n})^2 \rangle}{\langle |J_{\mathrm{spatial}}|^2 \rangle}, \tag{10}$$

where $f_\perp$ is the fraction of operation current flowing perpendicular to the plane. The admissibility projector becomes:

$$\Pi_{\mathrm{adm}}(x) = 1 - \frac{h_{\mathrm{lock}}(x)}{4}. \tag{11}$$

For a razor-thin settled disk, $h_{\mathrm{lock}} \to 1$ and $\Pi_{\mathrm{adm}} \to 3/4$ — recovering the derived endpoint. For a sphere, $h_{\mathrm{lock}} = 0$ and $\Pi_{\mathrm{adm}} = 1$. Real galaxies sit on a continuum between these limits based on how much normal-direction history has been written shut.

5 Source-force boundary and motion channel (R012–R013)

R012 showed that the direct $\nabla\chi \cdot J_{\mathrm{ops}}$ coupling evaluates to a boundary term for conserved worldlines. No local scalar fifth force is produced by S_{core} alone. Worldline motion must come through the settlement metric, the no-overwrite constraint, or event non-conservation (decay, scattering).

R013 tested a toy metric/settlement gradient channel and confirmed that S_{core} influences worldline motion through geometry rather than through a direct scalar force. The chain is:

> J_{ops} source geometry $\to$ repaired macro ledger action $\to$ settlement / metric geometry $\to$ future-write trajectory.

6 Frame, support, and effective sectors (R007, R006B, R008–R009)

Frame/holonomy (R007). Charge as worldline framing holonomy is gauge-compatible at the interface level: $D_\mu\theta = \partial_\mu\theta - qA_\mu$ with standard gauge transformations. This supports S_{frame} placement but does not derive α, charge quantization, or electroweak mixing.

Support domain (R006/R006B). A bare denominator cutoff admits the incorrect mode $\rho = 7/4$ (R006 failure). The full species formula requires primitive-anchor support-domain membership from $\{1, 1/2, 1/3\}$ plus binary/3+1 selection filters. The nine-mode seed is exactly reproduced.

Mass focusing (R008). V12.9's hierarchy-scale result (339,000$\times$ from geometry) is placed as an effective candidate functional. Pairwise hits remain null-downgraded. The sector status is EFFECTIVE_CANDIDATE.

Formation racetrack (R009). V12.10's formation program is placed as an effective kinetic layer. Heavy-first/light-last ordering, radiation double-burden, and three-class ecology are preserved. The sector status is EFFECTIVE_KINETIC_TOY.

7 Weak-field projection bridge (R014–R019)

The weak-field morphology bridge places the SPARC/λ connection downstream of source geometry and support admissibility. The support-lock scalar h_{lock} (upgraded in R024–R025) defines the interpolating function endpoint:

$$\lambda = \lambda_{\text{sphere}} - (\lambda_{\text{sphere}} - \lambda_{\text{disk}})\, h_{\text{lock}}, \tag{12}$$

with $\lambda_{\text{sphere}} = 4/3$ and $\lambda_{\text{disk}} = 3\pi/8$.

R016/R016B tested the morphology projection against synthetic bulge/disk configurations with null taxonomy and heavy noise controls. R017 hunted for the disk normalization κ. R018 derived $\kappa = 3/4$ from the 3+1 slot binary no-overwrite disk cell. R019 promoted this into the constraint stack.

The weak-field bridge is now formally placed: it is downstream of the same no-overwrite constraint that selects the nine particle modes.

8 Stress testing and h-lock upgrade (R021–R025)

R021B attempted a continuum constraint-density formulation for $S_{\text{NOW}}/S_{\text{support}}$. R021C built a multi-component P_{shape} operator for bulge/disk/bar morphology (patched in R021C_B).

R022B constructed a smooth admissibility functional. R022C stress-tested the P_{shape} v1 operator under noisy synthetic components.

R023B derived h-lock scalar source-geometry candidates. R023C ran a sensitivity/ablation audit on P_{shape} v1.

R023A required four iterations (A through D) to pass a sentence-level claim-boundary validation on the manuscript source. The failures were validation-design issues, not physics errors. R023A_D passed.

R024 upgraded the admissibility scalar to a support-lock tensor scalar $h_{\text{lock}} = q_{\text{flat}} \times c_{\text{oblate}}$, suppressing bar/prolate false positives. R024B then identified that the upstream no-overwrite question is planarity (σ_3/σ_2), not oblate coherence, and added the current-history correction $f_\perp$ to distinguish settled disks from transient pancakes.

R025 folded the upgraded h_{lock} into the full Master Action, producing a 14-page technical companion document.

R023 FINAL: READY_TO_WRITE_SFT_V12.11.

9 Sector certification summary

Sector	**Status**	**Certified by**
S_{core}	Action-ready core	R002
S_{hist}	Bridge-ready	R003, R005
S_{NOW}	Constraint-ready (upgraded)	R004, R018–R019, R024–R025
S_{frame}	Interface-ready	R007
S_{support}	Canonical constraint	R006B
$S_{\text{focus}}^{\text{eff}}$	Effective candidate	R008
$S_{\text{eff/thermal}}$	Effective kinetic toy	R009
P_{shape}/λ	Weak-field bridge	R014–R019, R024–R025

10 What is earned, conditional, and open

10.1 Earned

The repaired V11 core action survives all hard tests.

The worldline-to-current bridge connects V12's upgraded ontology to the old action without breaking it.

The χ-current coupling is a boundary term, not a fifth force.

No-overwrite is promoted from principle to constraint layer.

The disk admissibility factor $\kappa = 3/4$ derives from 3+1 slot binary no-overwrite structure.

The disk endpoint $\lambda_{\text{disk}} = 3\pi/8$ follows from raw chord times admissibility.

The nine-mode seed requires primitive-anchor support-domain membership (bare cutoffs fail).

Every V12 sector has explicit placement and certification status in the Master Action.

The smooth admissibility scalar h_{lock} is upgraded from plain flattening to σ_2-referenced planarity with current-history correction, providing a continuous interpolation between sphere ($\Pi = 1$) and disk ($\Pi = 3/4$) limits.

10.2 Not earned

Full continuum Euler–Lagrange closure.

Gauge coupling derivation (α, charge quantization, electroweak mixing).

Particle-to-mode mapping.

Native mass-action derivation.

Physical formation/cosmogenesis closure.

Full galaxy morphology solver or SPARC re-fit.

11 Conclusion

V12.11 shows that the repaired V11 Lagrangian does not need to be replaced. It needs to be recognized as the macro ledger core of a larger structure. The theory outgrew its original action

— not because the action was wrong, but because the ontology deepened. Particles are no longer featureless sources. They are maintained no-overwrite worldline histories with internal structure that generates the operation current, constrains the morphology projection, and opens the force, species, mass, and formation sectors.

The Master Action v1 is the first formal expression of SFT that places every post-V11 advancement explicitly. Some sectors are action-level. Some are constraint-level. Some are effective. Some are open. All are labeled honestly.

The most important sentence in V12.11 may be:

> *SFT currently has a complete repaired macro action, not yet a complete native microphysical action. V12.11 maps exactly where the boundary lies.*
>
> *The original Lagrangian made particles sources of the ledger. V12 requires particles to be represented as ledger-writing histories. V12.11 is the first action that knows the difference.*

Acknowledgments

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References

[1] S. Fosmark, *Solvency Field Theory V11: Master Framework*, Zenodo (2026).

[2] S. Fosmark, *SFT V12.4: Force-Sector and Electroweak Bridge*, Zenodo (2026).

[3] S. Fosmark, *SFT V12.7: Mechanical Filter Theorem Stack*, Zenodo (2026).

[4] S. Fosmark, *SFT V12.9: Mass as Geometric Focusing*, Zenodo (2026).

[5] S. Fosmark, *SFT V12.10: The Formation Racetrack*, Zenodo (2026).

A Gate rollup: R001–R023

Gate	Title	Contribution
R001	Action audit matrix	Sector table and classification for all V11.1–V12.10 advancements.
R002	Core survival	V11 action passes all hard tests. $V(0)=0$, $V'(0)=0$, $V''(0)>0$.
R003	Worldline current bridge	J^{μ}_{ops} as coarse-grained maintained-history current.
R004	No-overwrite admissibility	Written support locked. Arrow of time from constraint.
R005	Helical clock/phase	Compton phase Lorentz-compatible. No preferred frame.
R006	Support domain (FAIL)	Bare cutoff admits 7/4. Fixed in R006B.
R006B	Canonical support domain	Primitive-anchor R/C membership required. Nine-mode seed exact.
R007	Frame/holonomy interface	Charge as framing holonomy is gauge-compatible.
R008	Mass focusing placement	V12.9 hierarchy placed as effective candidate.
R009	Formation placement	V12.10 racetrack placed as effective kinetic toy.
R010	Master Action v0	Seven-sector stack assembled with status labels.
R011	Equation skeleton	Explicit mathematical form for each sector.
R012	Source-force boundary	χ-current coupling = boundary term, not fifth force.
R013	Settlement/motion bridge	Motion through metric geometry, not scalar gradient.
R014	Weak-field/SPARC bridge	λ morphology projection placed downstream.
R015	Projection operator	Candidate with open endpoint normalization.
R016/B	Morphology null controls	Synthetic bulge/disk with heavy noise and null taxonomy.
R017	κ derivation hunt	$\kappa = 3/4$ identified as structural candidate.
R018	Disk cell derivation	$\kappa = 3/4$ from 3+1 slot binary no-overwrite.
R019	κ promotion	Disk cell constraint promoted into S_{NOW}.
R020	Master Action v1	Full rollup with constraint-promoted projection bridge.
R021	Stress extensions	Paper outline, continuum attempt, multi-component P_{shape}.
R022	Stress tests	Formal note draft, smooth admissibility, noisy stress test.
R023	Final readiness	Sentence-level validation passed (R023A_D). Ready to write.
R024	Support-lock tensor	$h_{\mathrm{lock}} = q_{\mathrm{flat}} \times c_{\mathrm{oblate}}$ kills bar false positives.
R024B	σ_2 planarity lock	Upstream lock uses σ_3/σ_2. Current-history correction $f_{\perp}$ added.
R025	Master Lagrangian update	h_{lock} upgraded in full Master Action document.